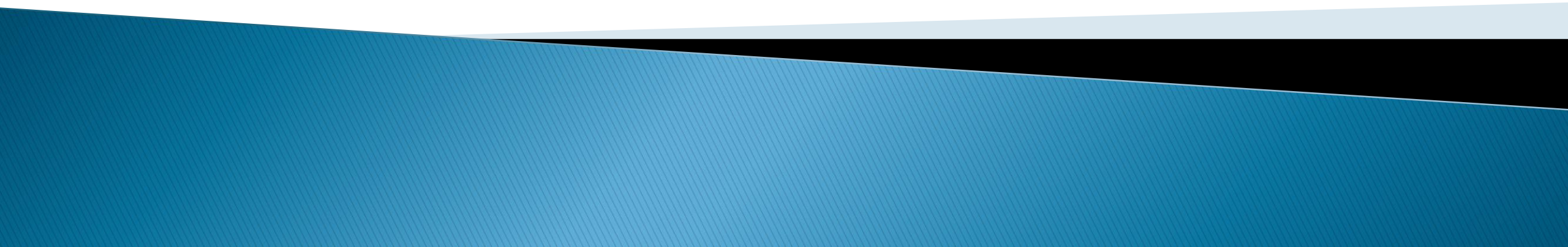
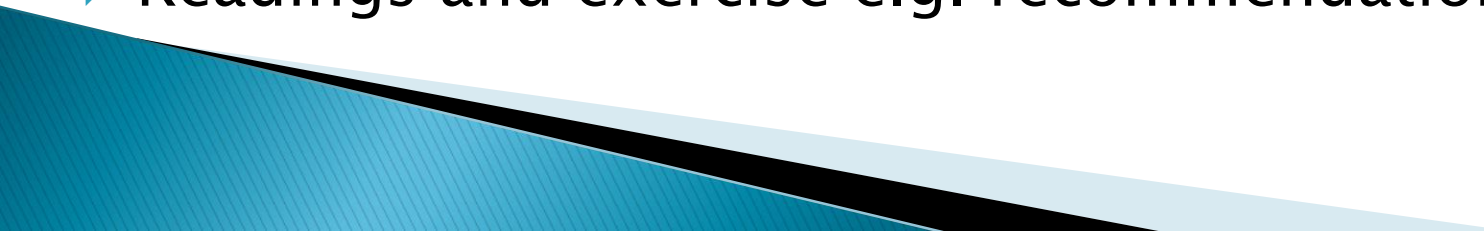


# Ethical issues in Responsible computing: AI

A brief overview



# So far...

- ▶ Introduction
  - ▶ Dimensions of ethical inquiry
    - Descriptive approaches to ethical inquiry
    - Normative approaches to ethical inquiry
  - ▶ Laws, codes vs principles
  - ▶ Ethical Questions at Different Levels of Social Organization
  - ▶ Various sources of norms & Why we need to reflect on the approaches to ethics and the sources of ethical norms.
  - ▶ Dilemmas
  - ▶ Ethical frameworks and fundamental ethical principles.
  - ▶ Design ethics
  - ▶ Codes of ethics
  - ▶ Readings and exercise e.g. recommendation engines, privacy
- 

# Next steps...

- ▶ Some specific topics and technologies
    - Recommendation engines
    - Big data and privacy
    - Algorithmic filtering (focus on newsfeeds and searches)
    - Big data and healthcare
  - ▶ Cybersecurity and some cyber related crimes
  - ▶ Social issues and computer science (digital misinformation etc.)
  - ▶ Digital divides, democracy and work
  - ▶ Intellectual property
- 

# Recommendation Engines, Fairness, and Transparency

Overview



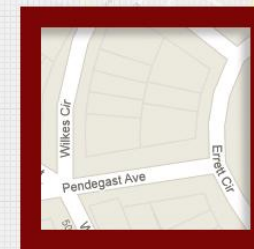
# Lecture Outline

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- ▶ Recommendation Engines
  - ▶ Fairness
  - ▶ Transparency and Opacity
  - ▶ Key Questions
- 



# Recommendation Engines



**PredPol**  
Predict Crime in **Real Time**

PredPol provides targeted, real-time crime prediction designed for and successfully tested by officers in the field.

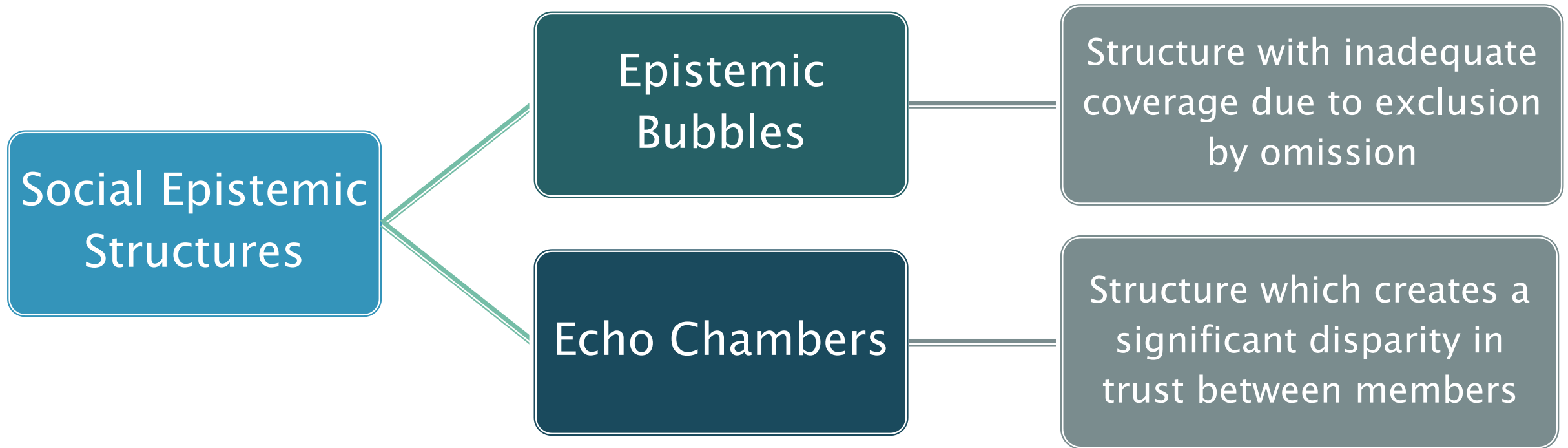


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# Epistemic Bubbles vs Echo chambers



# Topics and Technologies

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Recommendation  
Engines

Recidivism  
Prediction/Hiring  
Algorithms

Fairness

Predictive Policing

Respect & Individualized  
Reasoning

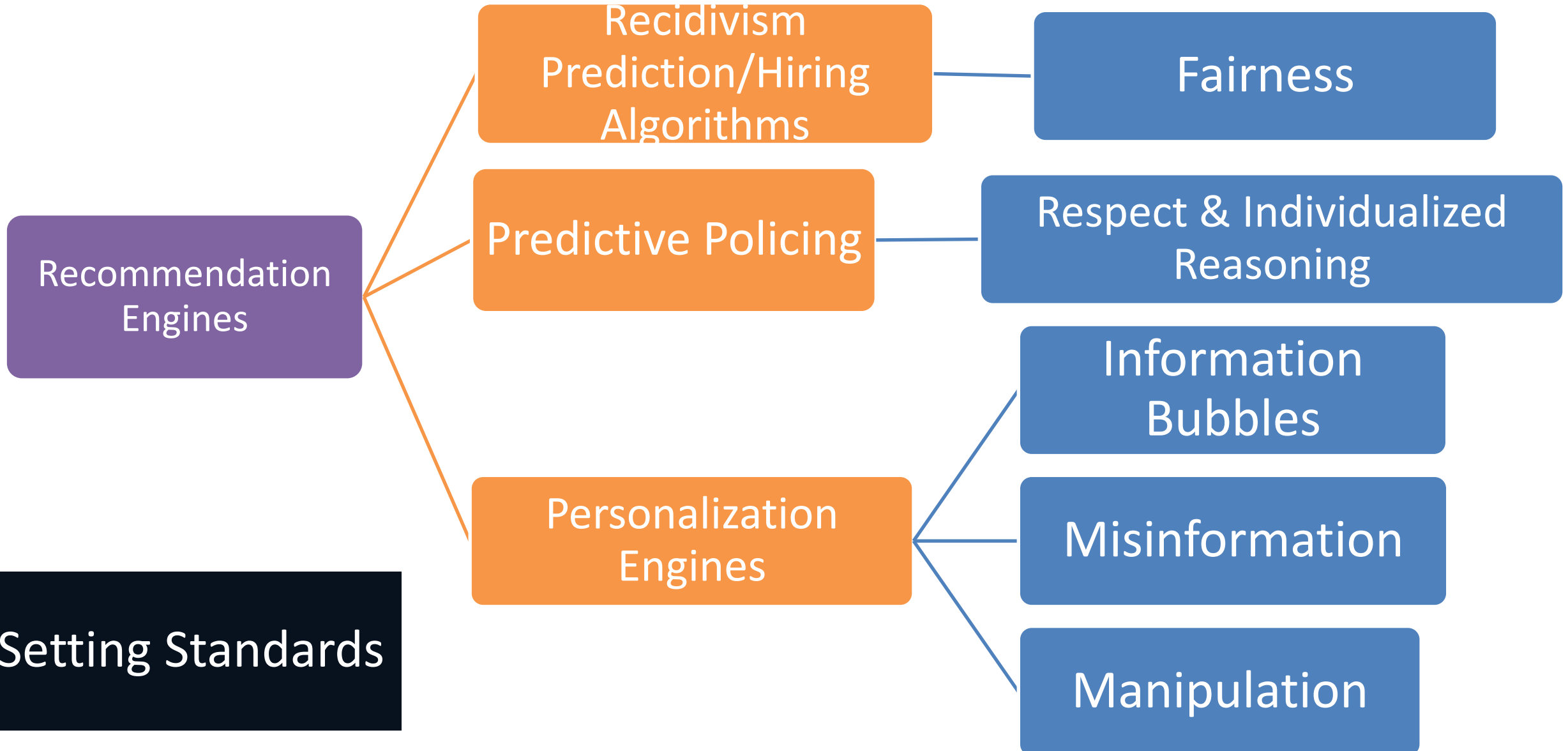
Personalization  
Engines

Information  
Bubbles

Misinformation

Manipulation

Setting Standards





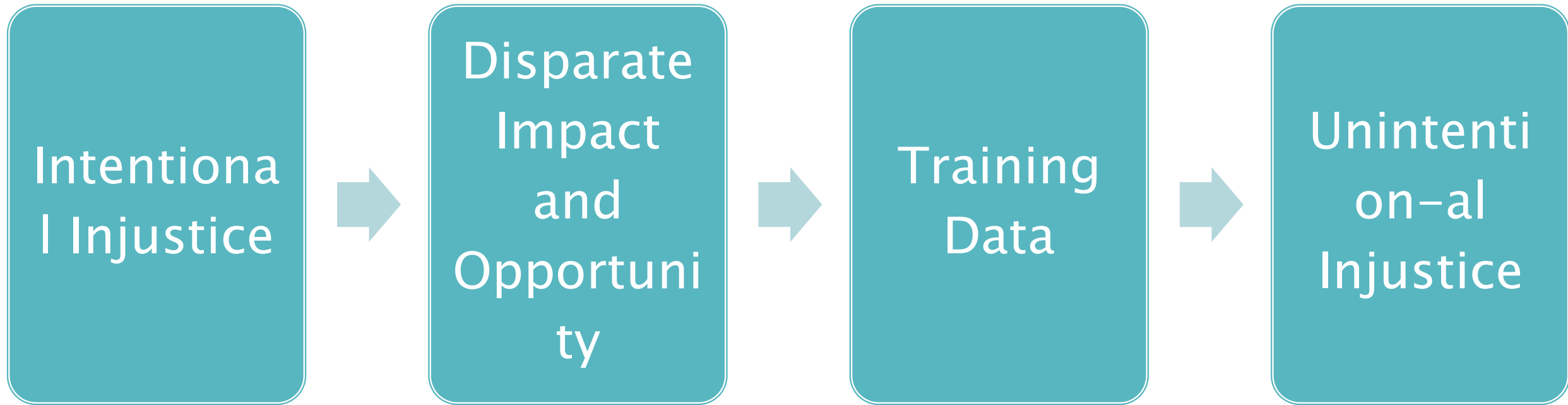
# Fairness

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# Fairness

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# Fairness

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Intentional  
Injustice



Disparate  
Impact and  
Opportunity



Training  
Data



Unintentional  
Injustice

Sexism



Impacts on  
Woman in the  
Workplace



Training  
Data



Algorithms  
disproportionatel  
y hire men  
independent of  
qualification

Racism



Impacts on  
criminal  
prosecution of  
BIPOC populations



Training  
Data



Algorithms  
disproportionately  
punish individuals  
from BIPOC  
populations